Modulus Functions & the Factor/Remainder Theorem

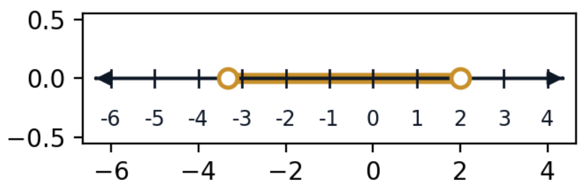
Evaluating and solving modulus equations

1 Solve $|2x - 3| = 7$.

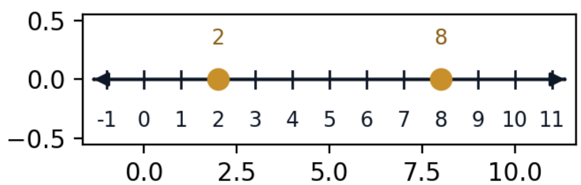
2 Solve $|x - 4| = |2x + 1|$.

Modulus inequalities

3 Solve the inequality $|3x + 2| < 8$, giving your answer as an interval.

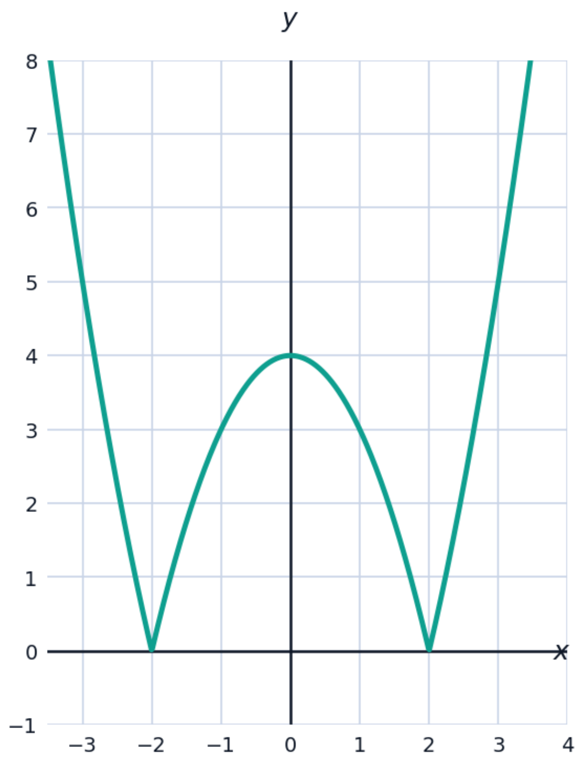


4 Solve $|x - 5| \geq 3$, giving your answer as a union of intervals.



Visualizing the graph of $y = |f(x)|$

5 The graph shows $y = |x^2 - 4|$.



a Explain how this graph relates to the graph of $y = x^2 - 4$, and identify which portion has been reflected.

b State the x-values where the graph touches the x-axis, and explain why the curve has sharp corners there rather than smooth turning points.

The factor theorem

6 Use the factor theorem to show that $(x - 2)$ is a factor of $f(x) = x^3 - 3x^2 - 4x + 12$, then factor $f(x)$ completely.

7 Find the value of k such that $(x + 3)$ is a factor of $f(x) = x^3 + kx^2 - 5x - 6$.

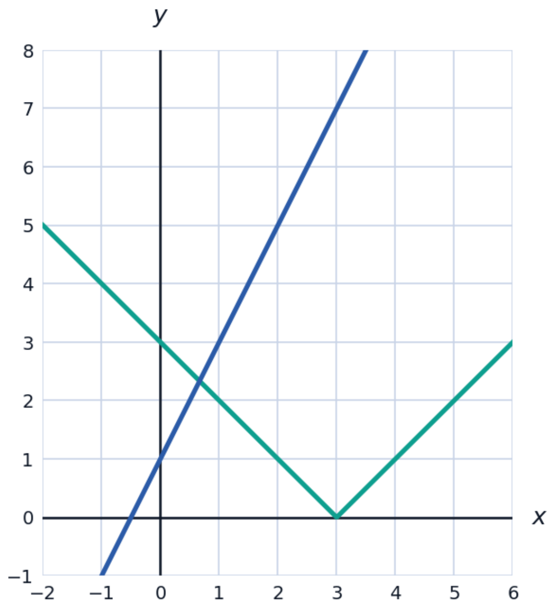
The remainder theorem

8 Find the remainder when $f(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $(x - 2)$.

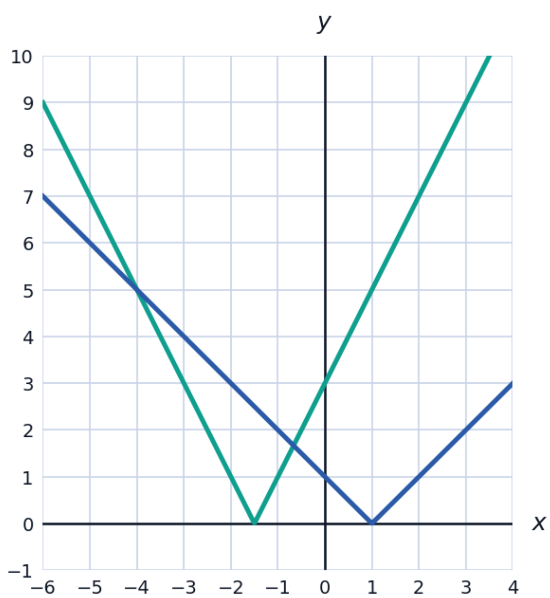
9 When $f(x) = x^3 + ax^2 + bx - 4$ is divided by $(x - 1)$ the remainder is 2, and when divided by $(x + 2)$ the remainder is -22. Find a and b.

Solving equations involving modulus functions graphically

10 Solve $|x-3| = 2x+1$, checking that each solution satisfies the original equation (since squaring or case-splitting can introduce extraneous solutions).



11 Solve the inequality $|2x+3| \leq |x-1|$ by squaring both sides (valid since both sides are non-negative).



Finding all zeros of a higher-degree polynomial

12 Given that $x = 1$ and $x = -1$ are both zeros of $f(x) = x^4 - 5x^3 + 5x^2 + 5x - 6$, find the remaining two zeros.

Using the remainder theorem to test multiple divisors at once

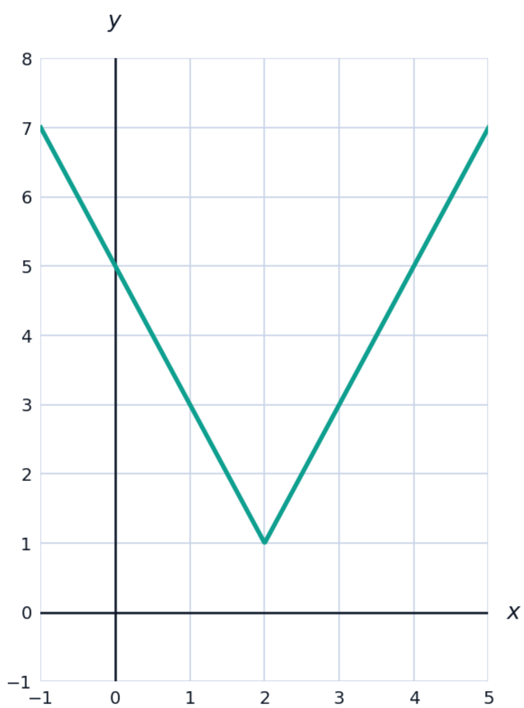
13 A polynomial $f(x)$ leaves remainder 5 when divided by $(x-2)$ and remainder -3 when divided by $(x+1)$. Find the remainder when $f(x)$ is divided by $(x-2)(x+1)$, given that the remainder must be of the form $ax+b$.

Solving polynomial equations with the factor theorem

14 Solve $2x^3 - x^2 - 13x - 6 = 0$ completely, given that $x = -2$ is one root.

Graphing modulus functions with restrictions

15 Sketch, in words, the key features of $y = |2x - 4| + 1$: state its minimum value, the x-value where it occurs, and its y-intercept.



Finding unknown coefficients using both theorems together

16 The polynomial $f(x) = x^3 + ax^2 + bx + 6$ has $(x-1)$ as a factor and leaves a remainder of -12 when divided by $(x+2)$. Find a and b.

17 Show that $f(x) = x^3 - 6x^2 + 11x - 6$ has three distinct real roots by testing $x = 1, 2, 3$ using the factor theorem, and write $f(x)$ in fully factored form.